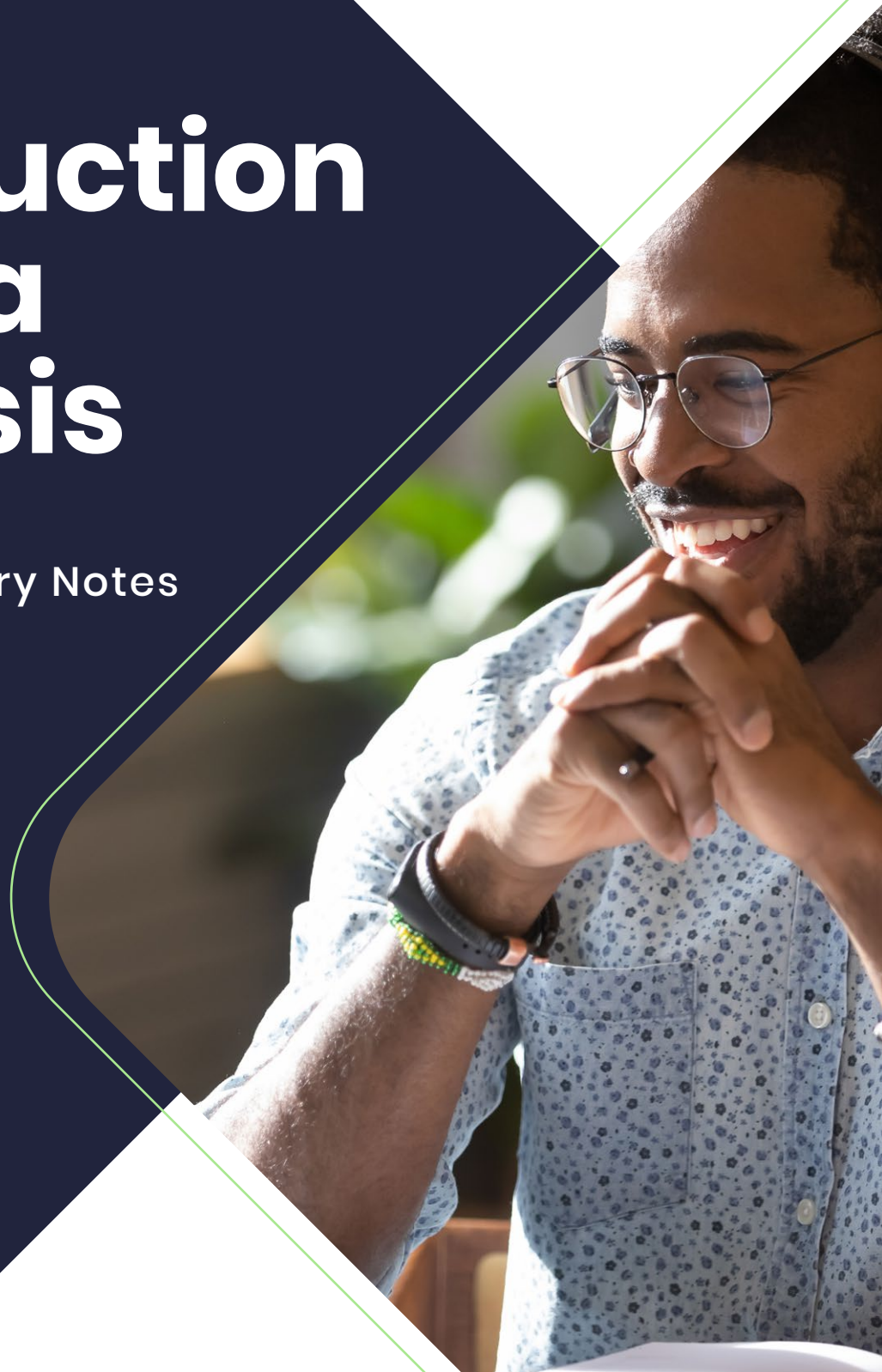


Diploma in Data
Analysis

Introduction to Data Analysis

Lesson 6: Summary Notes



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Lesson Objectives

- Hypothesis testing
- Types of errors
- Sample size

Lesson Introduction

On the titanic, Rose insisted that red wine should be served at slightly colder than room temperature to enhance the taste. Jack considered a hypothetical experiment to determine whether 100 passengers onboard could tell the difference in taste between the same wine served at various temperatures. In each test, the passenger would note whether the wine tasted good or bad to them. They used the results to determine whether Rose's initial belief was indeed correct. Do you think there was enough evidence in favour of Rose's initial belief?

Topics for this lesson include hypothesis testing and the concepts surrounding it, like Type 1 and 2 errors. You might have dealt with hypothesis testing, but might have found the concepts confusing. Hypothesis testing is not a difficult concept to understand, but unfortunately often gets presented in a way that is overwhelming and difficult to master. For that reason, we are going to break the concept down into bite-sized pieces to ensure you master the concept piece by piece. For the last topic of the lesson, we will rewind to the concept of sample size and enforce what it means to have a large enough sample.



Hypothesis testing

Hypothesis defined

According to the dictionary, a hypothesis is “a supposition or proposed explanation made on the basis of limited evidence as a starting point for further investigation.”

In other words, a hypothesis can be a question that is posed by us, the researcher. From that question, we want to make inferences about the population. Therefore, a hypothesis can be a claim or question that we want to test.

Estimator

Point and intervals estimates can be used to make inferences about the population which means that it is the process by which we acquire information and draw conclusions about populations from samples.

Parameters describe the population and sample statistics describe the samples. The sample statistics are almost always unknown and as a result, we take a random sample of the population to obtain the sample statistics we need. Estimation is one of the procedures we use to make inferences about the population. Hypothesis testing is the other.

Hypothesis testing steps

There are a lot of different cases that you have to consider when dealing with hypothesis testing, but for this lesson, we will first focus on hypothesis testing about the population mean with a sample size bigger than 30 and an approximately normally distributed population.

1. The first step in our test would be to establish the null and alternative hypothesis.
2. Thereafter, we define the critical value that defines our rejection and acceptance regions.
3. Thirdly, we calculate our test statistic.
4. Finally, we draw a conclusion by either accepting or rejecting the null hypothesis.

Step 1: Null and alternative hypothesis

The null hypothesis is the value we currently accept as true. We can denote this as H_0 . Now we want to determine if the null hypothesis is true and we test it using the alternative hypothesis. This is denoted by H_a .

When testing the hypothesis about the population mean, the null hypothesis would be equal to a certain value and the alternative hypothesis would show a difference to the value.

Example of hypothesis test

For example, you claim that a certain brand of chocolate bar's mean weight is not equal to the 100g it says it weighs on the wrapper.

The null hypothesis would be that the mean weight is equal to the value that is currently being accepted as true, 100g, and the alternative hypothesis is the claim that you want to test, that the mean weight is not equal to 100g. These two, the null and alternative hypothesis, are opposites of each other.

Methods of sampling

There are two possible outcomes to this example:

- Reject the null hypothesis or
- Fail to reject the null hypothesis

We do not accept the null hypothesis, because we are merely stating that we believe the null hypothesis to be more true than the alternative hypothesis for our current set of data.

The decision to reject or not reject is based on information contained in a sample taken from the population that we divide into an acceptance and a rejection region. The point dividing the regions is called the critical value. The normal distribution never changes its shape, so whether we have a sample with 40 observations or 50 observations, the distribution's shape stays the same.

Therefore, our critical values will always remain the same at the various confidence levels for the normal distribution:

- For 90% confidence level in the test, with an alpha value of 0,05, the critical values will be +/- 1,645
- For 95% confidence level in the test, with an alpha value of 0,025, the critical values will be +/- 1,96
- And for 99% confidence level in the test, with an alpha value of 0,01, the critical values will be +/- 2,575

The critical value, therefore, divides the sample space into a rejection and acceptance region. It creates a cut-off line so to speak.

If the sample statistic lies in the acceptance region, then H_0 is not rejected, or if the sample observation lies in the rejection region, then the null hypothesis is rejected.

Step 3: Test statistic

This lesson assumes a normal distribution for the sample and two-tailed test statistic. Therefore, our samples will be greater than 30 observations.

We calculate the test statistic from the sample data and use it to determine whether to reject or not reject the null hypothesis. A test statistic measures the degree of agreement between a sample of data and the null hypothesis.

The general formula for the test statistic in this course will be:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{hypothesized parameter}}{\text{standard error of the sample statistic}}$$

Use the variable z as the test statistic for a normal distribution where the population standard deviation is known.

The formula for the z test statistic is:

$$z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$$

Where

$\bar{X}$ = sample mean

μ = population mean

S = population standard deviation

$\sqrt{n}$ = sample size

Calculate where the test statistic lies and if it lies in the rejection region, you would reject the null hypothesis. The larger the value of the test statistic, the more likely it is that the observed difference reflects a true underlying difference (that is, a statistical significance). It is easy to see that statistical significance can arise with any combination of a large difference, or with less inherent variation, or with a larger sample size.

Example of critical value and regions

Let's revisit our chocolate example again.

You decided to go to the shop and sample 30 chocolate bars and calculate the average or mean value for the chocolate bars. The mean value is the test statistic.

Let's say that the chocolate bars have a mean weight of 99 g.

The level of confidence could be 95% in the random sample. This could also be stated in a different way by using alpha level of significance as 0,05. This is just a different way of stating the level of confidence.

Step 4: Conclusion

The last step would be to either reject the null hypothesis and accept the alternative hypothesis.

1. If the test statistic falls in the rejection region, we will reject the null hypothesis, as a result, we conclude that there is enough evidence to infer that the alternative hypothesis is true.
2. The other outcome would be to not reject the null hypothesis. If the test statistic does not fall in the rejection region, we do not reject the null hypothesis. We conclude that there is not enough evidence to infer that the alternative hypothesis is true.

Once again, we never state the conclusions with absolute certainty, and therefore we do not say that we accept the null hypothesis.

One sample hypothesis testing in Excel

Rose says that the average age for the passengers on the Titanic is 20. Jack says this is not true and that the average age is 25.

H_0 : mean age = 20

H_a : mean age $\neq$ 20

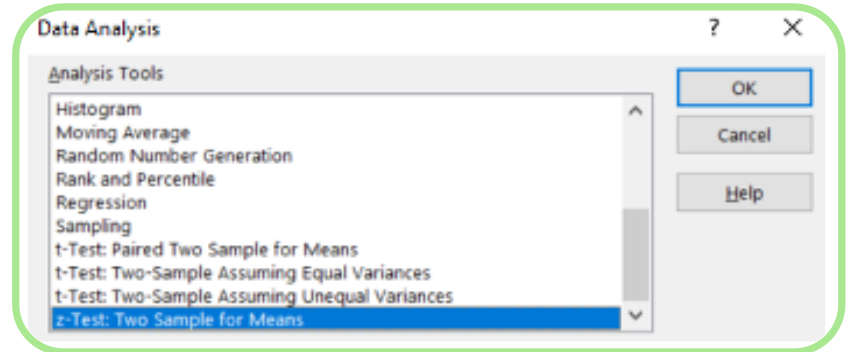
1. Create value to test variable against

NOTES

2. Select z-Test: Two Sample for Means from Data Analysis Toolpak

Age	Test
22	20
38	20
26	20
35	20
35	20
54	20
2	20
27	20
14	20
4	20
58	20
20	20
39	20
14	20

3. Input ranges and variances



z-Test: Two Sample for Means

Input

Variable 1 Range:

Variable 2 Range:

Hypothesized Mean Difference:

Variable 1 Variance (known):

Variable 2 Variance (known):

☐ Labels

Alpha:

Output options

☐ Output Range:

☒ New Worksheet Ply:

☐ New Workbook

Buttons: OK, Cancel, Help

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Type 1 and 2 errors

Hypothesis testing

When dealing with hypothesis testing, the results of the tests fall within a range of probabilities.

In essence, hypothesis testing is a procedure to compute a probability that reflects the strength of the evidence (based on a given sample) for rejecting the null hypothesis.

We previously determined this as an alpha value of 0,05 and called this our critical value.

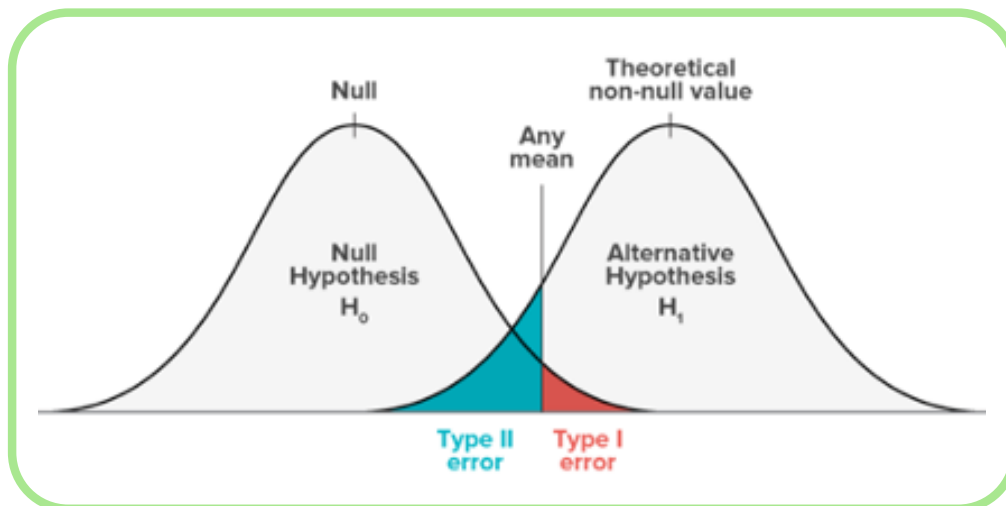
Alpha is a probability, therefore there is a 5% probability of incorrectly rejecting the null hypothesis when it is in fact true.

Errors

These mistakes are called Type 1 and Type 2 errors:

- A Type 1 error is when the null hypothesis is rejected when it should not have been.
- A Type 2 error is not rejecting the null hypothesis when it should be rejected.

We use the level of significance, our alpha value, to calculate the critical value and rejection region and this is our probability that we are willing to risk a type of error with.



To further explain how these errors affect our conclusion, we can look at the visualisation.

A Type 1 error, also known as a false positive, occurs when we incorrectly reject the null hypothesis. We can lower our risk of making this error, by using a lower value for alpha (the red area would become smaller), but that would then increase our chance of committing a Type 2 error as seen through this graph.

A Type 2 error, to recap, can also be known as a false negative.

Type 1 and 2 error example

Let's illustrate the types of errors with our chocolate example. If the mean sample weight of the 30 chocolate purchased was within statistically significant bounds to the mean weight of the weight on the wrapper, meaning that the sample mean falls within the 95% confidence interval, but we then reject the null hypothesis, we would make a Type 1 error.

Alternatively, If the mean sample weight of the 30 chocolate purchased was not within statistically significant bounds to the mean weight of the weight on the wrapper, meaning that the sample mean did not fall within the 95% confidence interval, but we then did not reject the null hypothesis, we would make a Type 2 error.

Example of Importance of errors

If you are to accuse the chocolate company of selling bars that weigh less than advertised, I want to be as certain as you can that you are not wasting their time or resources through a Type 1 error.

A Type 2 error might occur when you find there to be no difference in the mean sample weight from the expected, but in reality, the weight is different and an intervention is needed for all chocolate bars!



NOTES

Recap: Sample size

Let's revisit sample sizes. This is a complex topic to understand but is extremely important when designing a study that will lead to valid conclusions.

Cohen's seminal power analysis concluded that over half of published studies were insufficiently powered to result in statistically significance for the main hypothesis.

- Sample size refers to the amount of observations included in the sample.
- From the Central Limit Theorem we know that a distribution will be approximately normally distributed when the sample is bigger than 30 observations or data points. Typically, a sample size of 30 or more observations is sufficient for most distributions to tend to normal.



Statistical power of hypothesis test

Sample size is one of the factors impacting the power of the study.

A power of the study tells us the probability of correctly accepting the null hypothesis.

In other words, the power of the statistical test is the probability of the test to reject the null hypothesis when the null hypothesis is false.

The power says how likely we are of finding a statistically significant result from our results.

Statistical power

- The statistical power of a variable relates to the Type 2 error and reflects the probability of not committing a Type 2 error
- Mathematically, power is written as: $1 - \beta$.
- If the power is close to 1, the hypothesis will more likely detect false null hypothesis.
- An increase in our sample size, will give us greater power to detect differences.
- Other factors that influence power is our alpha level and the variance in the variable.

Example of statistical power

If we have a variable that has a power of 80%, it has a good chance of detecting statistically significant effects in the variable. In other words if we had a study that was conducted many times, having a power of 80% means that 80% of the time, we would detect a statistical significant difference in our results.

NOTES

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